Castles in the Air

Extended chapter commentary

CHAPTER 2
Beads and bracelets

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CHAPTER 2 COMMENTARY

Beads and bracelets

In Chapter 1, we looked at some of the 'bricks-and-mortar' ingredients of the mathematical edifice: the concepts of axiom, theorem and proof, and the idea of step-by-step deductions proceeding according to pure logic. In this chapter, I'd like to zoom out and look at the building a bit more architecturally: at how different theorems, concepts and methods typically fit together and contribute to something bigger. Along the way, we'll highlight some further aspects of the subject that any decent philosophy of mathematics will need to account for.

How does mathematics advance? Once you've proved a theorem, what can you use it for? What does it let you achieve? In what ways do mathematical theorems contribute to the general advance of human knowledge?

I'd like to illustrate some possible answers to these questions using the first theorem we met in this chapter: Fermat's so-called 'Little Theorem' (*not* to be confused with his celebrated 'Last Theorem', a different kettle of fish entirely which we'll come to later in the chapter). The Little Theorem was stated by Fermat in a letter to a friend in 1640, though the first published proof was given by Euler in 1736. Many proofs of this result are known; the ingenious bracelet-counting argument given in the story was found by Solomon Golomb in 1956.[1]

The theorem, as you will recall from the story, states that if p is any prime number and c is any Whole number, then p exactly divides $c^p - c$. So now that we have this theorem, what can we do that we couldn't do before?

I'd like to suggest three possible answers. The first of these is a rather obvious one: prove *more* theorems!

The power to prove

We have already noted the cumulative nature of mathematics: medallions can unlock new doors leading to further medallions. Fermat's Little Theorem is an excellent example of this: in fact, it's probably fair to say that this theorem represents the start of serious number theory. In the next chapter of our story, for instance, we shall see how it provides a crucial step in Emily's proof that there are infinitely many *evil* prime numbers. The concept of an evil

number will be explained in the next chapter. For now, it really doesn't matter what it means, but there's one aspect of the situation which it's convenient to touch on here.

There's a bit more to this example than meets the eye. After all, the *statement* 'There are infinitely many evil primes' doesn't seem to have anything to do with the idea of raising anything to a prime power p. In fact, one could perfectly well come up with this statement, and try to prove it, without ever having heard of Fermat's theorem. And it's rather hard to imagine that you would ever have arrived at Fermat's theorem *just* by trying to prove the infinitude of evil primes. Imagine that you are in the Castle some time in the 17th century, and have been trying without success to reach the 'Infinitude of Evil Primes' medallion which you suspect to reside in Turret A. (I don't know whether anyone *did* try this at this time; at any rate, no one succeeded.) Then one day, you learn that one Monsieur Fermat, pursuing quite different interests, has found a route to a remarkable new medallion in Turret B, some way off in a completely different direction. You then wonder whether this might be the key you need in order to get to A. After a bit of effort, you might discover that you are right. But the important point is that you would probably never have found the whole route for yourself just by trying to get to A: Turret B was simply too far out of the way for that. It wasn't until the Fermat medallion had been found by an explorer *on their way somewhere else* that the evil primes proof came within reach. Thus, theorems can give rise to other theorems, not just in the purely logical sense that they can be used to prove them, but in the practical sense that they bring them within reach of discovery.

This is maybe a slightly contrived example, but it illustrates the principle well enough. Indeed, mathematicians do sometimes voice the feeling that certain results are probably 'inaccessible' for the time being — that we won't be able to solve them by any amount of bashing our heads against them, but that further progress will just have to wait for a new idea coming from an unexpected direction.[2] Progress in mathematics is not always a matter of systematic, methodical exploration, but often comes about in indirect and serendipitous ways.

The power to calculate

Of course, mathematics would be something of a navel-gazing activity if mathematical theorems were only good for proving further theorems. But advances in mathematics let us do other things too: in particular, a theorem sometimes gives us the *ability to calculate* in ways that we couldn't before. And this aspect of mathematics often turns out to be of considerable value within what is sometimes known as 'the real world'.

In the story, we introduced the idea of an *engine*, a device for performing some kind of calculation mechanically. As medallions correspond to theorems, so engines correspond to what mathematicians call *algorithms*. An algorithm is, if you like, a set method or recipe for computing something, such as the method for long division that you learnt at school. Typically, it's a method that simply involves following certain fixed rules in a purely mechanical fashion — the sort of thing one could program a computer to do. A good algorithm will be one that works *efficiently* — that is, one that can really be used in practice, because it yields an answer after a relatively small number of calculation steps.

Now, the point is that theorems can often unlock the door to interesting new algorithms. Once again, we may turn to Fermat's theorem for an example. Remember that Fermat's theorem says something about the remainder when, for example, 2^p is divided by p. And as we saw in the story, there's a simple and efficient algorithm for computing what this remainder actually is. We don't need to know the details here,[3] but the crucial point is that the algorithm works efficiently even if p is a number of (let's say) 500 decimal digits. Although the number 2^p itself would then be far too big to be calculated in full, the *remainder* when this number is divided by p can actually be calculated within microseconds on a typical home computer. Combining this with Fermat's theorem, then, we have a pretty good test for whether a number p is prime: compute the above remainder using this efficient method, and just look and see whether it's 2 or not. Or, in place of 2, take any number c less than p, and test whether $c^p \div p$ leaves remainder c.

As we have seen, this 'Fermat test' isn't infallible: there are a sprinkling of numbers like 341 (known as *pseudoprimes* or *Fermat liars*) which pass the test, but which aren't really prime. Actually, these liars are pretty sparse: it's known that if you pick a 500-digit number at random, and use a randomly chosen base c for the test, the probability that you've picked a genuine prime number is around 1 in 1,200, whilst the probability that you've picked a Fermat liar is less than 1 in 10^{57}. So if your number passes the Fermat test, you can be pretty confident that it's a prime.[4] There are also some more sophisticated primality tests in the same spirit which are *guaranteed* to yield the right answer;[5] but in the meantime, even the humble Fermat test gives pretty good value for money.

What exactly is Fermat's theorem giving us here that we didn't have before? It's not simply that it gives an algorithm for testing primality. After all, we (and Emily) have always known one of those: just check in turn whether p is divisible by any smaller numbers! (We can of course stop once we reach the square root of p.) The point, rather, is that for a 500-digit number, this 'obvious' algorithm isn't remotely practicable, because the number of smaller numbers we'd have to check is utterly prohibitive — once again, even a computer the size of the universe could only make a small dent

in the task. However, by using a clever piece of mathematics, we're able to replace this obvious algorithm by a vastly superior one that really can be used on 500-digit numbers. This is what I mean by saying that mathematical theorems often bring with them the power to calculate.

But what, if anything, has this to do with 'real world'? Are any of these calculations actually of Practical Use? Does anyone in business or industry actually care about whether some gargantuan number is prime or not? Remarkably, the answer is a resounding 'Yes' — because large prime numbers lie at the very heart of modern cryptography and computer security. Next time you make a purchase on the Internet and supply your credit card details, this sensitive information will (I hope) be transmitted in an encoded form so that it doesn't fall into the wrong hands. Many different cryptographic methods are nowadays in use for such purposes, but by far the best-known is one called the *RSA cryptosystem*.

The RSA system (so named after Rivest, Shamir and Adleman, its three originators) is a simple but brilliant piece of number theory based on Fermat's Little Theorem, which uses prime numbers to encode messages in a safe and secure way. Again, we don't need to go into exactly how it works[6] — all that matters here is that it relies crucially on two ideas.

Firstly, RSA relies on the fact that prime numbers themselves are quite easy to find. If you want to let people send you encoded messages, you will need a way to generate a couple of large prime numbers, say p and q, which you then keep secret, although you do publicly reveal what $p \times q$ is. Now, primes aren't so very rare, so it's not hard to come up with two suitable primes as long as you have an efficient and practically reliable way of recognizing them. And this is where tests like the Fermat test come in. True enough, if you use a 'fallible' test like Fermat's, there's a tiny chance that you may hit on a 'liar' — though for business or commercial purposes this outside chance might be discounted, on the principle that it's far more likely that you and your entire company will be destroyed by a large meteorite before anyone has had time to send you a message. As the number theorist Henri Cohen put it, a large randomly chosen number that passes even the basic Fermat test can be considered an 'industrial grade prime'. But if you're not happy with this, you can always go to a little extra trouble and use one of the more rigorous primality tests. In either case, we have a striking example of a mathematical theorem yielding the power to calculate something that people really care about.

But the other idea on which RSA relies is also worth mentioning. Why is the code thought to be secure? This is because it's generally believed that breaking a 1000-digit number (say) into its prime factors is a very hard problem, beyond the reach of present computer technology. If someone could crack your number $p \times q$ to discover what p and q were, they would, quite literally, have cracked your code. At present, our trust in the security of RSA

rests on the assumption that no one knows how to do this. But if anyone secretly discovered a theorem that led to a spectacular new algorithm for factoring 1000-digit numbers, this would endow that person with a quite different, and more frightening, 'power to calculate'.

The power to predict

But I'd like to turn now to a third kind of ability that mathematics brings with it: the *ability to make predictions*.

There are many obvious examples of predictions made with the help of mathematics which concern actual events in the physical world: for instance, the time and place of the next solar eclipse. The ability to make predictions of this kind is, of course, fundamental to the deep and essential role that mathematics plays in the sciences. Of course, mathematics by itself can't conjure up such physical predictions out of thin air — it needs some scientific theories and other data to work on — but it's nonetheless true that the more mathematics we learn, the more we're able to predict about the behaviour of the physical world. And the better placed we are to design machines, too, because we can predict their behaviour before going to the trouble and expense of actually building them.

Many people have commented on the remarkable extent to which human mathematics seems to be capable of describing the physical world. Albert Einstein once wrote:

> How can it be that mathematics, being after all a product of human thought which is independent of experience, is so admirably appropriate to the objects of reality? Is human reason, then, without experience, merely by taking thought, able to fathom the properties of real things?[7]

We shall occasionally come back to this kind of question later on. But for the time being, I'd like to concentrate on the predictive power of mathematics not within physics, but within mathematics itself. In particular, mathematical theorems often let us predict what the results of certain calculations will be, in advance of actually doing them.

Yet again, we may appeal to Fermat's theorem as an example. Just now, we used this as a way of ascertaining that a number is probably prime, but now let's try looking through the other end of the telescope. Suppose we already know some that p is a prime: what does the theorem then tell us? Clearly, it *predicts* that if we pick any number c less than p, and compute the remainder on dividing c^p by p (using the efficient algorithm I mentioned earlier), the answer we observe will be just c. These sorts of calculational examples illustrate the predictive power of mathematics in a relatively pure form: the

predictions are not dependent on any particular scientific theory such as Newtonian mechanics or general relativity.

Why does this? What would ever be the point of *doing* a calculation if we could predict the answer in advance? Wouldn't that, in effect, just mean we had a more efficient algorithm for doing the calculation?

In fact, calculations whose outcomes are 'predictable in advance' arise more commonly than you might expect. Let's go back to the RSA cryptosystem for a moment. Imagine that you are sending me a secure message using this system. Your type your message on your smartphone, which encodes and transmits it, and my phone receives and decodes it. Now, it's predictable *to you* that the message I will read is exactly the message you typed. How do you know this? The mathematics of the RSA cryptosystem guarantees it, and in fact, what it boils down to is precisely the predictive power of Fermat's theorem as I've just described it. But of course, the decoding computation is still necessary if I am to read your message, even though the outcome of this computation is predictable in advance to you. So this is an example of how the predictive aspect of mathematics is important from the point of view of its real-world applications.

There's also a more philosophical reason for being interested in this idea of prediction. This will come to the fore in later chapters, but let me give you a glimpse of it here. As we saw in Chapter 1, one of our main philosophical questions is going to be: 'Which kinds of mathematical statements actually mean anything?' The idea is that 'predictive' statements like Fermat's theorem provide a kind of baseline: *at the very least*, any statement that makes a hard, testable prediction about what we'll see if we perform certain calculations must surely be said to have some 'meaning'. There are, of course, many other kinds of mathematical statements that *don't* make definite predictions in quite this way, and for these, the question of meaning might be more harder to address; but at least we've established a starting point.

The magic, so to speak, is that merely by 'taking thought', we seem to be able to make all sorts of mathematical predictions of this kind which are confirmed by actual calculation. For philosophical purposes, it seems better to focus on these purely mathematical predictions rather than on physical ones, so that the role of physical theories is kept to a minimum — we don't want to muddy the waters with a lot of worries over whether the physical theories themselves are valid.

But all of this is yet to come. In the next chapter, I'd like to zoom out even further and attempt something like an aerial view of mathematics on a much larger scale.

Endnotes

1. See S.W. Golomb, *Combinatorial proof of Fermat's 'Little' Theorem*, American Mathematical Monthly 63(10) (1956).
2. An easy-to-give example of a problem currently regarded as inaccessible (though admittedly a rather unnatural one) is the following: What is the 10^{50}th digit in the decimal expansion of π? This is far beyond the reach of direct calculation, but perhaps a clever proof could give us a handle on it? At present, no one has the foggiest idea how to begin to tackle such questions — but who knows what the mathematics of the next 1000 years might bring?
3. We can illustrate this algorithm easily enough by computing the remainder on dividing 2^{41} by 41. There are two ideas behind the algorithm. Firstly, we can avoid having to deal with any really big numbers, since at each stage of the calculation all we care about is the remainder on division by 41. So whenever we see a number bigger than 41, we can *reduce modulo 41* — that is, replace this number by its remainder: for example, $100 = 18 \pmod{41}$.
   Secondly, we can write 41 as a sum of powers of two: $41 = 1+8+32$. (This corresponds to writing 41 in binary notation as 101001.) So $2^{41} = 2^1 \times 2^8 \times 2^{32}$. Now, these special 'powers of powers of two' can be quickly computed (mod 41) like this:

$$2^1 = 2$$
$$2^2 = 4$$
$$2^4 = (2^2)^2 = 4^2 = 16$$
$$2^8 = (2^4)^2 = 16^2 = 256 = 10 \pmod{41}$$
$$2^{16} = (2^8)^2 = 10^2 = 100 = 18 \pmod{41}$$
$$2^{32} = (2^{16})^2 = 18^2 = 324 = 37 \pmod{41}$$

   We can now just pick out the powers of powers of two that we're interested in, and multiply them together:

$$2^{41} = 2^1 \times 2^8 \times 2^{32} = 2 \times 10 \times 37 = 740 = 2 \text{ (mod. 41)}$$

   Since 41 is prime, this is as predicted by Fermat's theorem.
   The point to notice is that in this calculation we never need to do anything *41 times*. Rather, we are doing various things 6 times, because 6 is the number of digits of 41 written in binary. This means the method is still practical for very large numbers: for example, a 500-digit decimal number will have about 1700 binary digits, so the calculation will (roughly speaking) involve doing various simple things 1700 times, which is chicken feed for a modern computer.
4. See S.H. Kim and C. Pomerance, *The Probability that a Random Probable Prime is Composite*, Mathematics of Computation 53(188), 1989.
5. See, for example, H. Cohen and H.W. Lenstra, *Primality testing and Jacobi sums*, Mathematics of Computation 42(165), 1980. A remarkable *polynomial*

time deterministic primality test was discovered in 2002 by Agrawal, Kayal and Saxena.

6. I shall resist the temptation to explain it here; it would take a page or so to do so. At the time of writing, there's an excellent Wikipedia article on 'RSA cryptosystem' which explains the ideas thoroughly and carefully, and includes a proof of correctness using Fermat's Little Theorem.
7. In A. Einstein, *Geometry and experience*. Lecture at the Prussian Academy of Sciences, Berlin, Methuen & Co. (1922).